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/*****
* AUTHOR:      CYRIL ANDRE B. DURANGO
* CLASS:      CPE 3101L - INTRODUCTION TO HDL
* SCHEDULE:   Group 4 | FRIDAY | 7:30 AM - 10:30 AM
*-----
* PROJECT FILE TITLE: Problem_A.v
*-----
*****/

module Problem_A
(
    input      [3:0] Month_in,
    input      LeapYear_in,

    output reg  NumDays28_out,
    output reg  NumDays29_out,
    output reg  NumDays30_out,
    output reg  NumDays31_out
);

    always @(*)
    begin
        casez (Month_in,LeapYear_in)
            5'b0001? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
            5'b00100 : begin NumDays28_out = 1'b1 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b0 ; end
            5'b00101 : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b1 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b0 ; end
            5'b0011? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
            5'b0100? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b0 ; end
            5'b0101? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b1 ; NumDays31_out = 1'b0 ; end
            5'b0110? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b0 ; end
            5'b0110? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b1 ; NumDays31_out = 1'b0 ; end
            5'b1000? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
            5'b1001? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
            5'b1010? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b1 ; NumDays31_out = 1'b0 ; end
            5'b1011? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
            5'b1100? : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b1 ; NumDays31_out = 1'b0 ; end
            default : begin NumDays28_out = 1'b0 ; NumDays29_out = 1'b0 ; NumDays30_out = 1'b0 ; NumDays31_out = 1'b1 ; end
        endcase
    end
end

```

Quartus Prime Lite Edition - Z:/HDL Program Files/Problem_A/Problem_A - Problem_A

File Edit View Project Assignments Processing Tools Window Help

Project Navigator Hierarchy Table of Contents Flow Summary

MAX 10 10M50DAF484C8G

Problem_A

Flow Summary

Flow Status: Successful - Fri Dec 06 08:50:09 2024

Quartus Prime Version: 20.1.1 Build 720 11/11/2020 SJ Lite Edition

Revision Name: Problem_A

Top-level Entity Name: Problem_A

Family: MAX 10

Device: 10M50DAF484C8G

Timing Model: Final

Total logic elements: 5

Total registers: 0

Total pins: 9

Total virtual pins: 0

Total memory bits: 0

Embedded Multiplier 9-bit elements: 0

Total PLLs: 0

URM blocks: 0

ADC blocks: 0

Tasks

Task

Compile Design

Analysis & Synthesis

Fitter (Place & Route)

Assembler (Generate program in)

Timing Analysis

EDA Netlist Writer

Edit Settings

Program Device (Open Programmed)

Messages

System (5) Processing (12)

12021 Found 1 design units, including 1 entities, in source file problem_a.v

12021 Found 1 design units, including 1 entities, in source file tb_problem_a.v

12127 Elaborating entity "Problem_A" for the top level hierarchy

286030 Timing-Driven synthesis is running

16010 generating hard_block partition "hard_block:auto_generated_inst"

21057 implemented 14 device resources after synthesis - the final resource count might be different

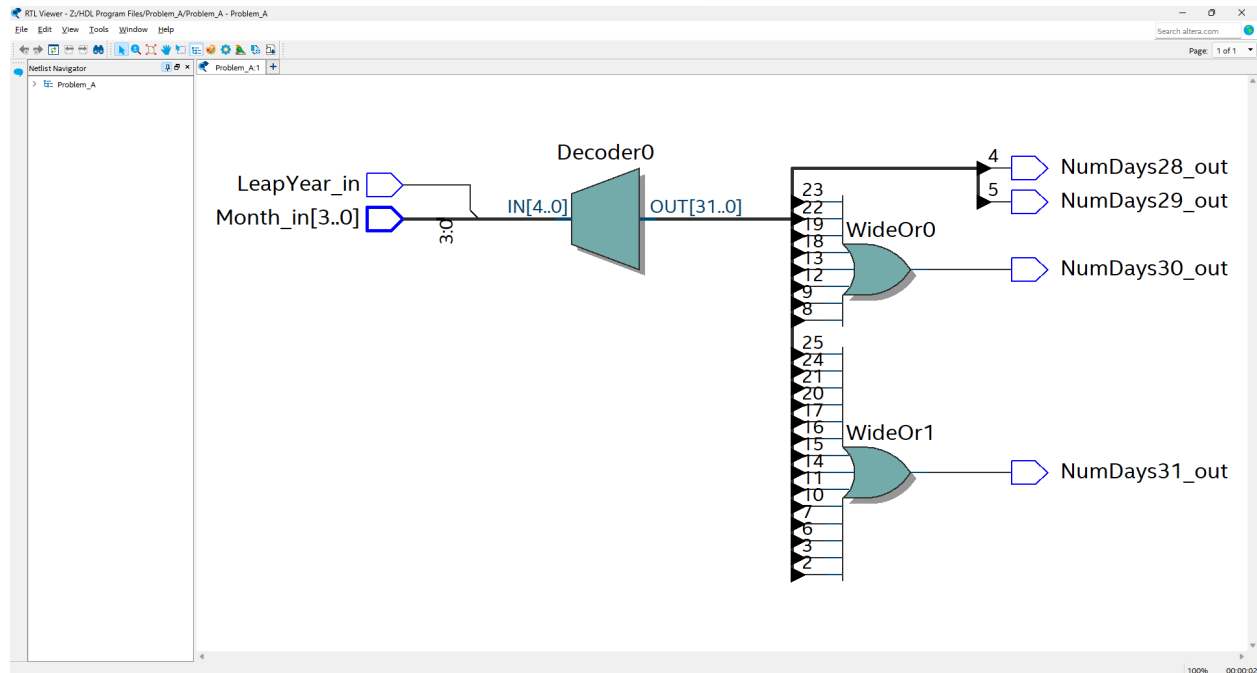
Quartus Prime Analysis & Synthesis was successful. 0 errors; 1 warning

Snipping Tool

Snip copied to clipboard

Select here to mark up and share the image

100% 00:00:08



Problem_A.v

Compilation Report - Problem_A

tb_Problem_A.v

```

10 module tb_Problem_A();
11
12     reg      [3:0] Month_in;
13     reg      LeapYear_in;
14
15     wire     NumDays28_out;
16     wire     NumDays29_out;
17     wire     NumDays30_out;
18     wire     NumDays31_out;
19
20
21     Problem_A uut(Month_in, LeapYear_in, NumDays28_out, NumDays29_out, NumDays30_out, NumDays31_out);
22     initial begin
23         LeapYear_in = 0 ; Month_in = 4'd0; #10
24         Month_in = 4'd1; #10
25         Month_in = 4'd2; #10
26         Month_in = 4'd3; #10
27         Month_in = 4'd4; #10
28         Month_in = 4'd5; #10
29         Month_in = 4'd6; #10
30         Month_in = 4'd7; #10
31         Month_in = 4'd8; #10
32         Month_in = 4'd9; #10
33         Month_in = 4'd10; #10
34         Month_in = 4'd11; #10
35         Month_in = 4'd12; #10
36         Month_in = 4'd13; #10
37         Month_in = 4'd14; #10
38         Month_in = 4'd15; #10
39
40         LeapYear_in = 1 ; Month_in = 4'd0; #10
41         Month_in = 4'd1; #10
42         Month_in = 4'd2; #10
43         Month_in = 4'd3; #10
44         Month_in = 4'd4; #10
45         Month_in = 4'd5; #10
46         Month_in = 4'd6; #10
47         Month_in = 4'd7; #10
48         Month_in = 4'd8; #10
49         Month_in = 4'd9; #10
50         Month_in = 4'd10; #10
51         Month_in = 4'd11; #10
52         Month_in = 4'd12; #10
53         Month_in = 4'd13; #10
54         Month_in = 4'd14; #10
55         Month_in = 4'd15; #10
56     |
57     $stop;
58 end
59

```

